NLP Coursework 3: Analysing Customer Reviews using NLP

Amazon Beauty Reviews Dataset: 7,990 valid reviews | positive = 3,996 (50.0%) | negative = 3,994 (50.0%)

(a) Rationale for Two-Stream Preprocessing Design

Two token streams were retained for distinct modeling objectives:

• **POS-filtered stream (NOUN and ADJ only, lemmatised):** Used for TF-IDF vectorisation and LDA topic modelling. In consumer reviews, topical themes are anchored by nouns (e.g. *brush, skin, wig, scent*) and sentiment is conveyed by descriptive adjectives (e.g. *soft, thin, great, awful*). Excluding verbs and grammatical stopwords drastically reduces vocabulary dimensionality and prevents syntactic noise from distorting topic distributions.

• **Lightly cleaned stream (all content words, no POS filter):** Used for Word2Vec continuous training. Distributed embeddings rely on sliding context windows (Skip-gram/CBOW), where preserving verbs, adverbs, and modifiers maintains local co-occurrence context and semantic nuance.

• **NER extraction:** Named Entity Recognition runs on the original unfiltered text to ensure entity boundaries and capitalization are preserved without restriction from preprocessing.

(b) Top 5 LDA Topics with Human-Readable Labels and Sentiment Split

Models with k in {5, 7, 10} were evaluated using Gensim's c_v coherence; **k = 5 was selected (c_v = 0.501)**.

Topic	Human-Readable Label	Top Representative Terms	Reviews	% Pos	% Neg
1	Makeup Tools & Application	skin, product, face, money, good	1,784	47.8%	52.2%
2	Eye & Face Makeup	hair, quality, good, wig, great	1,757	45.4%	54.6%
3	Nail Care & Quality	color, great, nail, small, size	1,682	54.9%	45.1%
4	Skincare & Cleansers	eye, lash, little, product, time	1,083	46.7%	53.3%
5	Hair & Wig Styling	product, good, brush, smell, scent	1,684	54.5%	45.5%

(c) Three-Row Classifier Comparison Table (Held-Out Test Split = 20%)

Classifiers trained on TF-IDF features evaluated on a stratified 20% test split (random_state=42):

Classifier	Accuracy	F1-Score	ROC-AUC
MultinomialNB	0.7716	0.7821	0.8522
BernoulliNB	0.7610	0.7561	0.8459
Logistic Regression	0.7610	0.7548	0.8548

(d) Insight Linking NER Entities to Sentiment

• **Brand Sentiment Insight:** Among the top mentioned brands, **Braun (ORG)** and **Norelco (ORG)** received the highest proportion of negative feedback, with Braun having a mean sentiment of **0.167 (83.3% negative)** and Norelco at **0.286 (71.4% negative)**. Negative reviews predominantly complain of replacement blade dullness, battery life, and compatibility problems.

• **Geographic Origin:** Mentions of **China (GPE)** similarly exhibit low sentiment (**0.167**, 83.3% negative), with customers expressing frustration over counterfeit parts or prolonged delivery.

• **Lexical Validation:** TF-IDF pairwise similarity confirms same-label mean = 0.0160; cross-label mean = 0.0131; same-label > cross-label = True.